Connor Voelk

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connorvoelk@gmail.com Seattle, WA

Software engineer with 4+ years at AWS and Tesla, building production GenAI, cloud, and IoT systems end to end, from distributed backends and mobile SDKs to the React interfaces people use every day.

Experience

Amazon Web Services (AWS) Seattle, WA

Cloud Application Developer *January 2024 – July 2026*

Designed and built production GenAI and IoT systems for enterprise customers, from architecture to launch.

Agentic AI Platform – Fortune 50 Healthcare Company – $1.9M Contract Value

- Enabled thousands of internal users to deploy autonomous agents for cross-functional API tasks by architecting an enterprise platform on LangChain, AWS Bedrock AgentCore, and MCP, running on Amazon EKS.
- Accelerated knowledge discovery across millions of data points by building a GraphRAG system on Amazon Neptune, Redis, and Textract that maps semantic relationships for retrieval by other GenAI products.
- Partnered with the Bedrock AgentCore service team as an early adopter, reporting gaps found in production.

Mobile Telemetry SDK – Market-Leading Satellite Provider – $1.5M Contract Value

- Led a team of 3 engineers building a Kotlin Android SDK from the ground up on dozens of system APIs, which automakers integrated into their apps and Android Auto for location sharing and emergency assistance.
- Reduced messaging costs 3.5x and achieved sub-50ms latency by replacing HTTP APIs with a pub/sub messaging layer over MQTT on AWS IoT Core.

Software Engineer, Delivery Consultant *July 2022 – January 2024*

Built full-stack web applications on React, GraphQL, and AWS backends for enterprise media customers.

ML-Powered Forecasting Tool – Major National Media Network – $2.0M Contract Value

- Increased ad-slot sales 15% on average with a simulation tool that let non-technical users model viewership scenarios in real time, visualizing demographic data for multi-million dollar scheduling decisions.
- Built the viewership forecasting model in Amazon SageMaker (feature engineering, training, inference), served through a React, Redux, and AppSync GraphQL app on DynamoDB.

Multi-Destination Live-Stream Orchestrator – Large Social Media Company – $1.4M Contract Value

- Saved a $1.4M contract by taking over a failing engagement, rebuilding the UI and backend in 4 weeks.
- Consolidated 8 streaming destinations into one AWS Elemental interface, cutting broadcast hardware overhead.

Tesla Palo Alto, CA

Software Engineer Intern *January – June 2021*

- Automated real-time HIL test data ingestion with a Python/Kafka pipeline, replacing manual reporting.
- Built a centralized React/Flask app for live sensor visualization and sub-second queries over legacy test data.

Education

University of Washington Seattle, WA

Bachelor of Science in Informatics, Specialization in Human Computer Interaction *2022*

Dean's List Scholarship, CSE UbiComp Lab Research Assistant, iSchool Teaching Assistant

Skills

Languages & Frameworks: TypeScript / JavaScript, Python, Java, Kotlin, SQL, Bash, GraphQL (Apollo), React.js, Redux, Node.js, Flask, LangChain, Jest, Playwright, Android, IaC (CDK, Terraform)

Tools & Technologies: AWS (Bedrock AgentCore, SageMaker, EKS, Neptune, DynamoDB, AppSync, IoT Core, Elemental), PostgreSQL, Redis, Kafka, Docker, Kubernetes, CI/CD, GraphRAG, MCP, REST, MQTT, Figma

AWS Certifications: Solutions Architect, Developer, Machine Learning Engineer (2026); AI Practitioner (2025)